

Recombinant Methioninase Selectively Eliminates Cancer Cells Co-cultured With Normal Fibroblasts Indicating the High-Precision Efficacy of Targeting Methionine Addiction of Cancer

BYUNG MO KANG^{1,2,3}, QINGHONG HAN¹, SHUKUAN LI¹, JIN SOO KIM^{1,2}, KOHEI MIZUTA^{1,2}, YOHEI ASANO^{1,2}, YUTA MIYASHI^{1,2}, MICHAEL BOUVET² and ROBERT M. HOFFMAN^{1,2}

¹AntiCancer Inc., San Diego, CA, U.S.A.;

²Department of Surgery, University of California, San Diego, CA, U.S.A.;

³Department of Surgery, Chuncheon Sacred Heart Hospital, Hallym University College of Medicine, Chuncheon, Republic of Korea

Abstract

Background/Aim: Methionine addiction is a fundamental and general hallmark of cancer and is targeted by methionine restriction. The aim of the present study was to examine whether methionine restriction can eliminate cancer cells co-cultured with normal cells that remain healthy.

Materials and Methods: The IC₅₀ for recombinant methioninase (rMETase) was determined for HC116 colon-cancer cells using the WST-8 viability assay. HCT-116 cells and Hs27 human normal skin fibroblasts were co-cultured and treated with rMETase at the IC₅₀ concentration for HCT-116 cells. Cell morphology and viability were monitored over 16 days via phase-contrast microscopy.

Results: The IC₅₀ for rMETase was 0.46 U/ml for HCT-116 cancer cells. In the untreated control co-cultures, the HCT-116 colon cancer cells overgrew Hs27 normal fibroblasts from day 2. In contrast, rMETase-treated cultures showed elimination of HCT-116 cells by day 8, while Hs27 fibroblasts remained viable and proliferative throughout the experiment.

Conclusion: The differential sensitivity of cancer and normal cells to rMETase at 0.46 U/ml was so extensive that cancer cells were essentially eliminated from the co-cultures while the normal cells remained viable and proliferating. The difference in sensitivity of normal and cancer cells growing together suggests that rMETase selectively and precisely targets methionine addiction of cancer and can be a safe and effective clinical anti-cancer agent.

Keywords: Methionine addiction, Hoffman effect, methioninase, cancer cells, normal cells, co-culture, selective elimination.



Robert M. Hoffman, AntiCancer Inc., 7917 Ostrow St., San Diego, CA 92111, U.S.A. Tel: +1 6198852284, e-mail: meishale@gmail.com

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Introduction

Methionine addiction is a fundamental hallmark of cancer termed the Hoffman effect (1). In several head-to-head comparative studies using PET imaging of cancer patients with [^{18}F]fludeoxyglucose (FDG) and [^{11}C]methionine, the Hoffman effect of methionine addiction has been shown to be generally stronger than glucose addiction of cancer cells termed the Warburg effect (2-4).

Numerous studies have focused on using recombinant methioninase (rMETase), an enzyme that degrades methionine, to effectively target methionine-addicted cancer cells (1, 5, 6). However, whether rMETase can selectively target cancer cells without affecting normal cells is not clearly understood. To address this question in the present study, we conducted a co-culture experiment of cancer and normal cells followed by rMETase treatment aiming to examine the differential sensitivity of cancer and normal cells to rMETase, and to demonstrate directly that the methionine addiction of cancer cells is so extensive that cancer cells can be eliminated from co-cultures with normal cells, while normal cells remain alive and can proliferate.

Materials and Methods

Cell culture. Human colorectal cancer HCT-116 cells and Hs27 normal dermal fibroblasts were obtained from the American Type Culture Collection (ATCC, Manassas, VA, USA). Both cell lines were maintained in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and penicillin/streptomycin at 37°C in a humidified atmosphere with 5% CO_2 .

Recombinant methioninase production. Recombinant L-methionine α -deamino- γ -mercaptomethane-lyase (rMETase), an enzyme that catalyzes the breakdown of methionine, was produced by fermenting genetically-engineered *Escherichia coli* containing the *methioninase* gene, originally derived from *Pseudomonas putida* (7). Production and

purification processes (7) were conducted at AntiCancer Inc. (San Diego, CA, USA).

Determination of the IC_{50} of rMETase on HCT-116 colon cancer cells. To determine the IC_{50} of rMETase on HCT-116 cells, 1,250 cells per well were seeded in 96-well plates. The following day, rMETase was added to six replicate wells at concentrations of 0, 0.125, 0.25, 0.5, 1.0, 2.0, 4.0, and 8.0 U/ml. After three days, cell viability was assessed using the WST-8 reagent from the Cell Counting Kit-8 (CCK-8) (Dojindo Laboratory, Kumamoto, Japan). The optical density (OD) at 450 nm was measured with a spectrophotometer (Sunrise™, Tecan Inc. Mannedorf, Switzerland) after addition of the WST-8 reagent to the cell culture (5). The IC_{50} was calculated from the dose-response curve.

Growth comparison of HCT-116 and Hs27 cells cultured alone with and without rMETase. To compare the growth of HCT-116 and Hs27 cells in response to rMETase, both cell lines were seeded in five 96-well plates, with 1,250 cells per well. Two rows (six wells per row) were assigned to each cell line. Two days after seeding, one row was treated with the rMETase IC_{50} for HCT-116 cells, and the other was replenished with fresh medium (no rMETase). On days 0, 2, 4, 6, and 8 post-treatment, cell viability was measured using the CCK-8 kit, and the OD at 450 nm was recorded with a spectrophotometer. The average OD from six replicate wells was used to estimate cell number at each time point.

Microscopic analysis of co-cultured HCT-116 and Hs27 cells with and without rMETase. To observe the efficacy of rMETase in co-culture, HCT-116 and Hs27 cells were seeded together in 6-well plates at 3.75×10^4 cells each (total 7.5×10^4 cells/well). Two days after seeding, the HCT-116 IC_{50} for rMETase was added to the treatment wells, while control wells received no rMETase. Phase-contrast microscopy was used to monitor growth status and cell morphology, and images were acquired using the IX71 inverted microscope (Olympus Corporation, Tokyo, Japan).

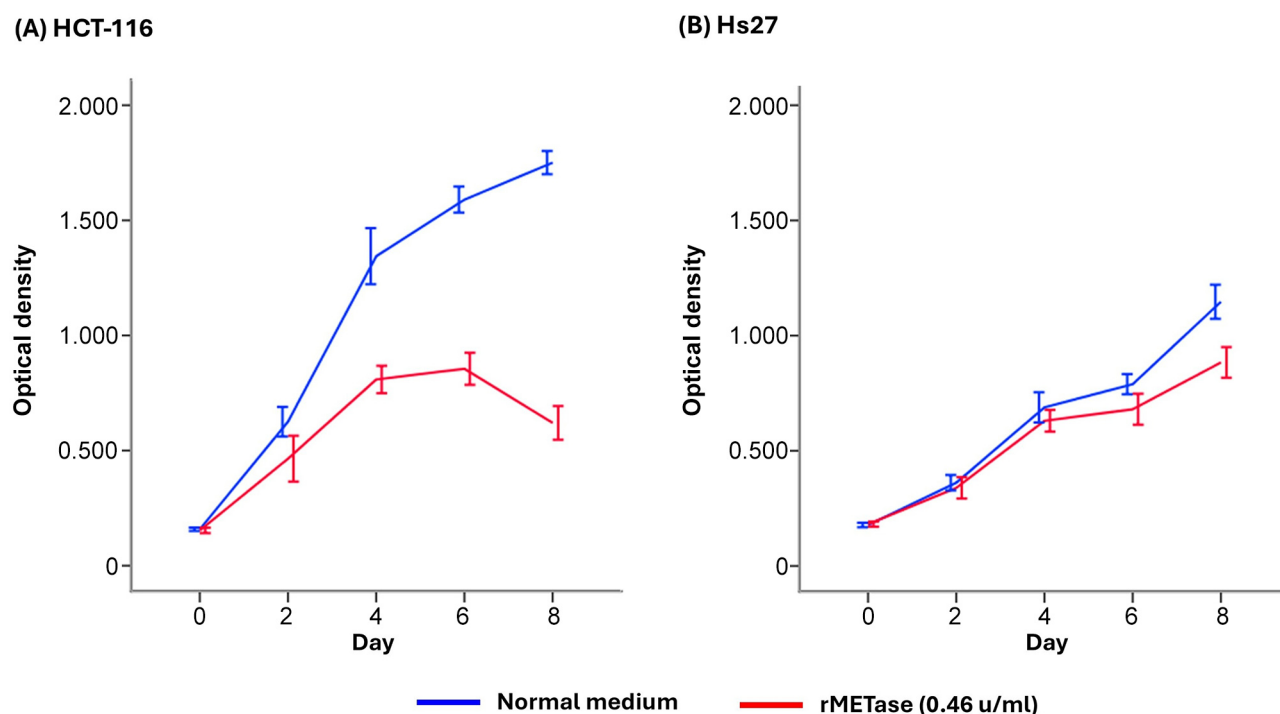


Figure 1. Growth of HCT-116 and Hs27 cells with or without rMETase treatment. Cell viability was assessed with the CCK-8 assay over 8 days. rMETase (0.46 U/ml) inhibited HCT-116 cell growth, while Hs27 cells showed minimal inhibition. Data represent the mean optical density (OD) at 450 nm after addition of the WST-8 reagent of six replicate wells per group at each time point. Please see the Materials and Methods for details.

Results

Efficacy of rMETase on HCT-116 and Hs27 cell growth. The IC_{50} of rMETase on HCT-116 cell growth was 0.46 U/ml. As shown in Figure 1, 0.46 U/ml of rMETase significantly suppressed HCT-116 cell growth. Cells in untreated wells proliferated steadily, while those treated with rMETase reached a plateau between days 4 and 6 and declined by day 8, indicating growth arrest or loss of viability. In contrast, Hs27 cells showed a minimal response to rMETase, with both treated and untreated groups displaying and comparable growth, and only slight reduction was observed in the treated group at later time points.

Effect of rMETase on HCT-116 and Hs27 cells in co-cultures compared to the untreated co-cultures. Phase-contrast microscopy, at 40-fold magnification, showed morphological differences between co-cultures with and without rMETase

(Figure 2). In the absence of rMETase, HCT-116 cells began dominating the co-culture from day 2, rapidly proliferating and overtaking the co-culture. Hs27 fibroblasts appeared as linear arrangements squeezed between cancer cell clusters, gradually diminishing and becoming barely visible by day 16 (Figure 2A). In contrast, rMETase-treated co-cultures showed sustained Hs27 growth and reduction of HCT-116 cells, which exhibited morphological changes consistent with cell death. By day 8, viable HCT-116 cells were no longer observable, while Hs27 cells remained healthy and proliferative (Figure 2B).

At 100-fold magnification on day 14 (Figure 3), untreated co-cultures displayed dense HCT-116 overgrowth, with Hs27 fibroblasts limited to sparse linear arrangements. In rMETase-treated co-cultures, few HCT-116 cells remained, appearing non-viable, while Hs27 fibroblasts predominated. Mitotic Hs27 cells were also observed (black arrows).

Discussion

The present results show an extensive differential sensitivity of cancer and normal cells to rMETase such that cancer cells were selectively eliminated from co-cultures of cancer and normal cells. This difference in sensitivity of cancer and normal cells to methionine restriction is due to the methionine addiction of cancer cells (5, 8, 9, 11).

Forty years previously we co-cultured normal and cancer cells in homocysteine-containing, methionine-free medium. The addition of chemotherapy drugs was necessary to eliminate the cancer cells in homocysteine-containing, methionine-free medium while maintaining viable proliferating normal cells (10). The present results indicate that rMETase has a much stronger selective effect against cancer cells compared to methionine-free, homocysteine-containing medium.

Methionine restriction is due to overuse of methionine by cancer cells. Although cancer cells make normal or greater amounts of methionine than normal cells they still need external methionine at high levels (11-14). The excess requirement of cancer cells for methionine is due, at least in part, to excess transmethylation reactions in cancer cells including histone lysine methylation (13, 15-18).

Based on these findings, the present study provides direct visual and quantitative evidence that rMETase alone can reverse the growth advantage that methionine-addicted cancer cells have over normal cells. In co-cultures, rMETase rapidly shifted the population dynamics: methionine-addicted HCT-116 cells plateaued and underwent cell death, while methionine-independent Hs27 fibroblasts continued to proliferate and eventually dominated the culture. This indicates that methionine depletion by rMETase not only inhibits the growth of cancer cells but also selectively spares the proliferation of normal cells.

The selective elimination of HCT-116 cells by rMETase became progressively more pronounced from day 8, consistent with the "methionine-depletion catastrophe" previously described in our earlier study (5). In phase-contrast microscopy of co-cultures on day 14, HCT-116 cells overwhelmingly dominated in co-cultures without rMETase,

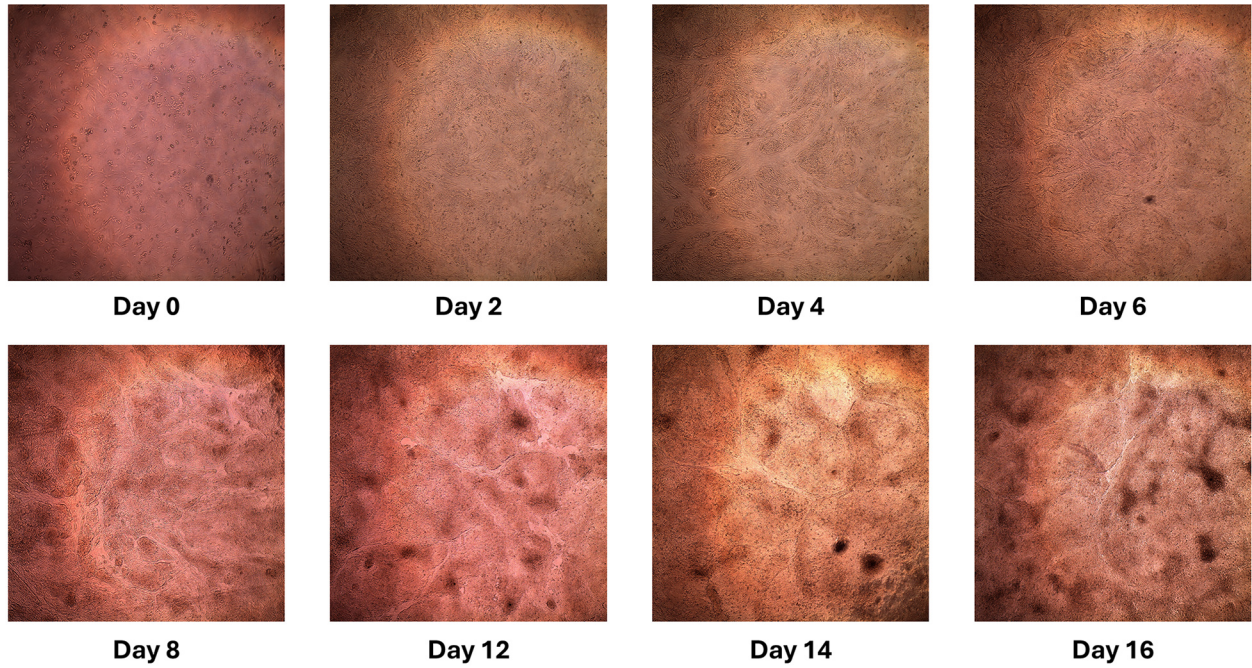
showing extensive overgrowth, whereas in rMETase-treated co-cultures, Hs27 fibroblasts predominated, with visible mitotic figures, and viable HCT-116 cells were scarcely observed (Figure 3). These results demonstrate that rMETase-induced enzymatic methionine depletion is more effective than dietary restriction strategies reported in earlier studies (19, 20).

On a biochemical basis, the increased sensitivity of cancer cells to methionine depletion induced by rMETase may reflect the elevated rate of transmethylation in cancer cells (17, 18), which supports the epigenetic changes required for tumor growth. rMETase reduces extracellular methionine to very low levels, which in turn lowers the production of S adenosylmethionine (SAM) in cancer cells (21), a key molecule used in transmethylation. This limits important methylation-dependent functions in tumors, such as histone modifications (*e.g.*, H3K4 and H3K9 trimethylation) that promote tumor growth and maintain genomic stability (8, 13, 15, 16). In contrast, normal fibroblasts are less affected by methionine depletion because they can utilize endogenously-synthesized methionine and have adequate levels of methionine use for transmethylation (17, 18). Normal as well as cancer cells can regenerate methionine from homocysteine *via* remethylation, a process that primarily depends on the folate cycle and vitamin B₁₂-dependent methionine synthase (11, 12). However, cancer cells still exhibit a critical dependence on exogenous methionine.

The present study confirms the methionine addiction of HCT-116 colon cancer cells and demonstrates the potential of rMETase as a highly selective and safe anti-cancer agent. Its ability to selectively eliminate cancer cells while preserving co-cultured normal fibroblasts within the same microenvironment supports its translational feasibility and highlights its favorable safety profile. Unlike conventional chemotherapeutic agents that often harm rapidly dividing normal cells, rMETase targets a specific metabolic vulnerability found only in cancer cells, thereby minimizing off-target toxicity.

Furthermore, our previous findings suggest that rMETase may be particularly effective in combination with

(A) Normal medium



(B) rMETase (0.46u/ml) treatment

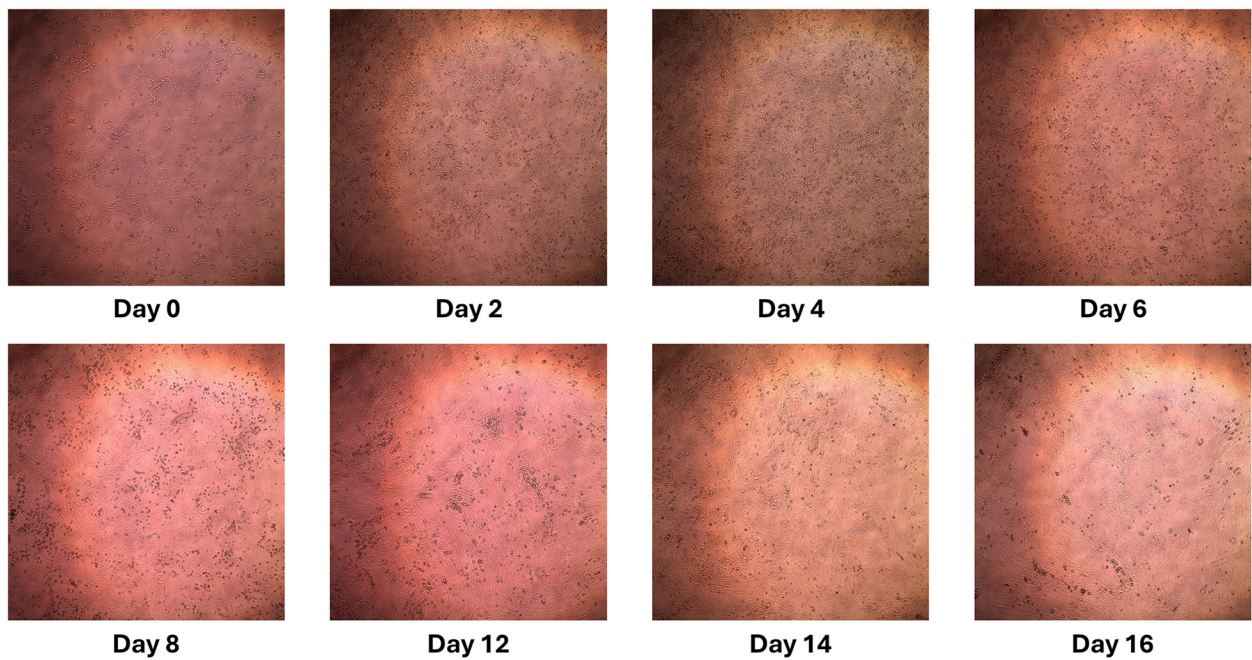
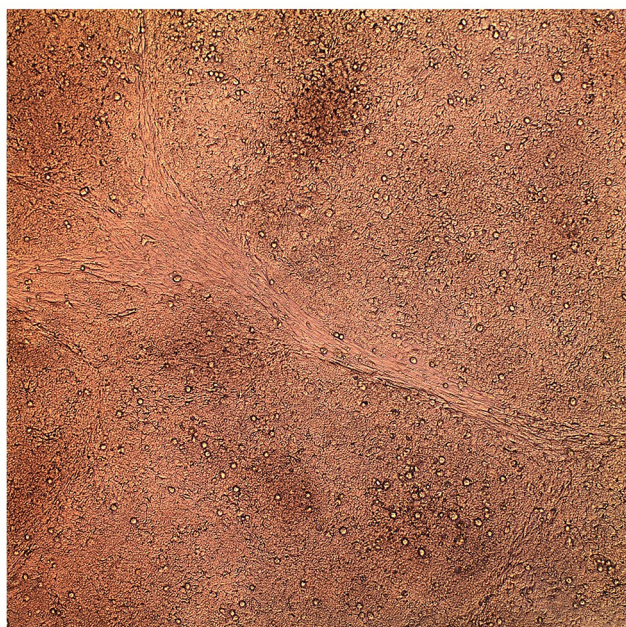


Figure 2. Phase-contrast microscopic images of co-cultured HCT-116 and Hs27 cells with and without rMETase treatment (40-fold magnification). (A) Without rMETase, HCT-116 cells rapidly overgrew Hs27 fibroblasts, which appeared as linear arrangements and nearly disappeared by day 16. (B) With 0.46 U/ml rMETase (IC_{50} for HCT-116 cells), Hs27 cells remained proliferative and dominant, while HCT-116 cells showed signs of cell death and were no longer observed by day 8. Please see the Materials and Methods for details.

(A) Normal medium



(B) rMETase (0.46u/ml) treatment

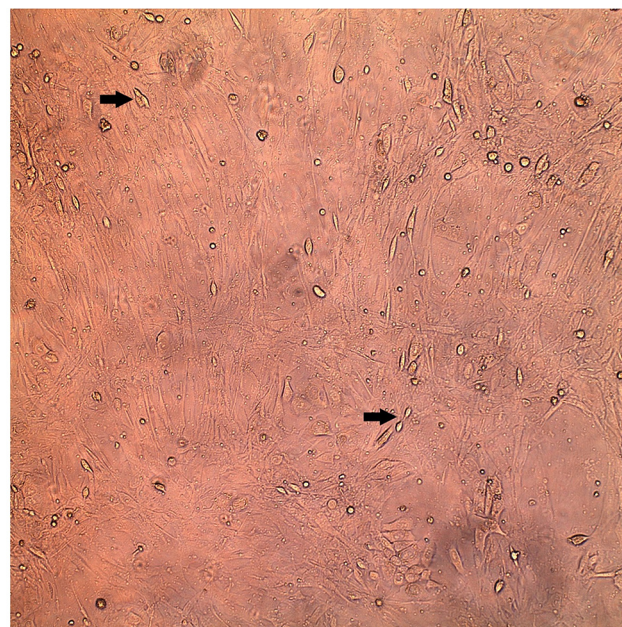


Figure 3. Phase-contrast microscopic images of co-cultured HCT-116 and Hs27 cells on day 14 with and without rMETase treatment (100-fold magnification). (A) Without rMETase, HCT-116 cells densely covered the surface, with Hs27 fibroblasts remaining only as sparse linear traces. (B) With rMETase, few non-viable HCT-116 cells were observed, while healthy, proliferating Hs27 fibroblasts predominated. Mitotic Hs27 cells are indicated by black arrows. Please see the Materials and Methods for details.

other therapies. Previous studies have demonstrated that methionine depletion by rMETase can enhance the efficacy of standard chemotherapeutic agents and immune checkpoint inhibitors (1, 23, 24). By depleting extracellular methionine, rMETase may sensitize methionine-addicted tumors to both cytotoxic chemotherapy and immune-based therapies, potentially enabling dose reductions and minimizing adverse effects. Building on these findings, we plan to compare the responses of cancer and normal cells in co-culture using methionine-free, homocysteine-supplemented media to complete media with rMETase treatment. This will further clarify their differential sensitivity to methionine restriction.

Conclusion

The differential sensitivity of cancer and normal cells to rMETase at 0.46 U/ml was so extensive that cancer cells were effectively eliminated from the co-culture, while

normal fibroblasts remained viable and proliferative. This selective effect underscores the potential of rMETase as a precise and cancer-specific therapeutic agent. Its methionine-depleting action may enhance the effectiveness of existing chemo- and immunotherapies, since it targets methionine addiction, which is tightly linked to malignancy (1, 9, 13, 25, 26, 27), and shows the clinical potential of methionine restriction (19, 20, 28-30).

Conflicts of Interest

All Authors have no conflicts of interest or financial ties to disclose related to this study.

Authors' Contributions

BMK and RMH conceived and designed the study. QH was responsible for the production of rMETase. BMK performed all experiments and drafted the manuscript.

RMH provided critical revisions. JSK, KM, YA, YM, and MB contributed to the critical review of the final manuscript.

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Artificial Intelligence (AI) Disclosure

No artificial intelligence (AI) tools, including large language models or machine learning software, were used in the preparation, analysis, or presentation of this manuscript.

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